



FALL SEMESTER 2025-26

BEVD301P

ASIC Design Lab

Lab TASK 5

Physical Verification and RC extraction

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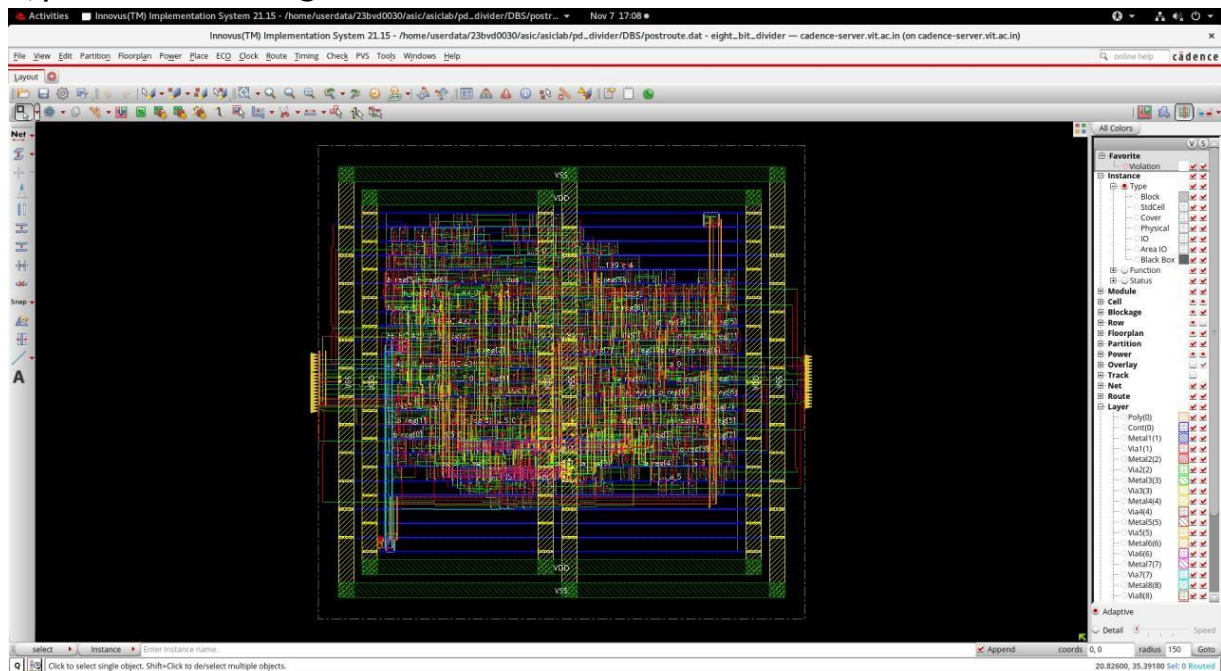
Slot: L5+L6

Date:7/11/2025

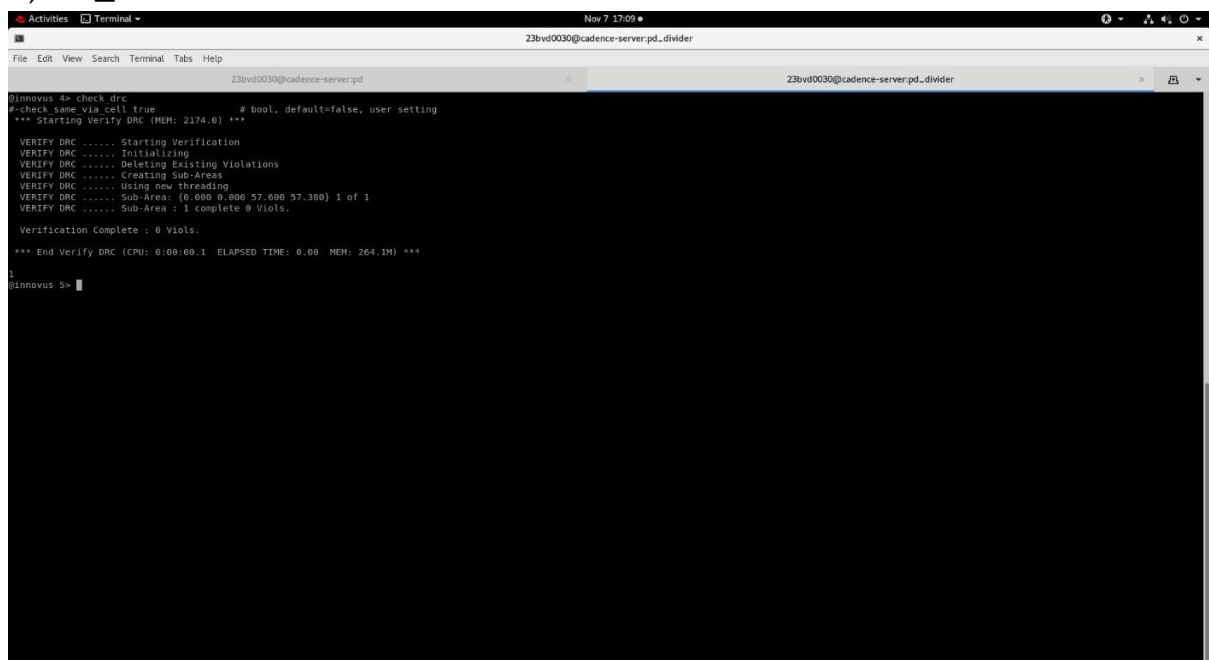
Aim:

Perform STA on the digital design using Cadence Innovus and Tempus to verify the timing integrity of a post-layout design. The goal is to extract correct parasitic data in SPEF format, apply appropriate timing constraints in SDC format, analyze reports of timing, find setup and hold violations, and perform timing optimization, such as buffer insertion, cell sizing, or refinement of constraints to ensure that all timing violations have been resolved and that timing closure has been achieved.

1)post_route design



2)drc_violations



3)check_connectivity

Activities Terminal Nov 7 17:49

23bvd0030@cadence-server:pd_divider

File Edit View Search Terminal Tabs Help

23bvd0030@cadence-server:pd 23bvd0030@cadence-server:pd_divider

```
@innovus 7> check connectivity
VERIFY_CONNECTIVITY use new engine.

***** Start: VERIFY CONNECTIVITY *****
Start Time: Fri Nov 7 17:16:33 2025

Design Name: eight bit divider
Database Units: 2000
Design Boundary: (0.0000, 0.0000) (57.6000, 57.3000)
Error Limit = 1000; Warning Limit = 50
Check all nets

Begin Summary
  Found no problems or warnings.
End Summary

End Time: Fri Nov 7 17:16:33 2025
Time Elapsed: 0:00:00.0

***** End: VERIFY CONNECTIVITY *****
Verification Complete : 0 Viols, 0 Wrngs.
(CPU Time: 0:00:00.0 MEM: 0.000M)
```

1
@innovus 8> █

4)rc_extraction

```

Activities Terminal Nov 7 17:10
23bv0030@cadence-server:pd_divider
File Edit View Search Terminal Tabs Help
23bv0030@cadence-server:pd 23bv0030@cadence-server:pd_divider

does NOT contain MetalFill data.
TechFile
does NOT contain WEE data.
does NOT contain Erosion data t=f( density )
does NOT contain R(w) data.
does NOT contain R(w, s) data.
does NOT contain TC(w) data.
does NOT contain T/B enlargement data.
does contain Floating Metal Fill models.

INFO (EXTIQUANTUS-287) : Capacitance Models Information :-
ICECAPS models are not available.
RCgen models are available.
This Quantus session uses RCgen models.

INFO (EXTIQUANTUS-286) : RCs effects computed for this session include :-
MetalFill : OFF
WEE Effects : n/a
Erosion Effects : n/a t=f( density )
T/B Enlargements : n/a
R(w) Effects : n/a
R(w,s) Effects : n/a
TC(w) Effects : n/a
Some effects indicate n/a because of unavailability of relevant input data
(or) requested to be off (and/or) usage of older Icecaps models.

INFO (EXTHPY-103) : Extracting capacitance values.

INFO (EXTHPY-267) :
DESIGN :
DEF/OA DIEAREA BBOX : 2147483647 2147483647 -2147483648
DEF/OA DIEAREA : 2147483648
DEF/OA UNITS : 10000
FINAL SCALE FACTOR : 1

ESTIMATED COMPRESSED DEF SIZE: 0
TILE SIZE : 50x50 sqm
TILE COUNT : 2
CLUSTER SIZE : 257
CAPDB PARTITIONS : 16

INFO (EXTHPY-104) : 0%

INFO (EXTHPY-104) : 100%
*****
Extraction of Geometries DONE (NETS: 490 CPU Time: 0:00:00.3 Real Time: 0:00:00.0 MEM: 2438.094M)

Parasitic Network Creation STARTED
*****
Number of Extracted Resistors : 5818
Number of Extracted Ground Caps : 5516
Number of Extracted Coupling Caps : 312
```

5)divder_spfef.spfef file

```
divder_spfef.spfef
~/.asiclab/pd_divider

Open  Save  --  X

*SPEF "IEEE 1481-1998"
*DESIGN "eight_bit_divider"
*DATE "Wed Nov 5 22:48:59 2025"
*VENDOR "Cadence Design Systems, Inc."
*PROGRAM "Innovus"
*VERSION "21.15-s110 1"
*DESIGN_FLOW "COUPLING C" "PIN_CAP NONE" "NAME_SCOPE LOCAL"
*DIVIDER /
*DELIMITER :
*BUS DELIMITER []
*T_UNIT 1 NS
*C_UNIT 1 PF
*R_UNIT 1 OHM
*L_UNIT 1 HENRY

// MIMC spfef file for corner 'rccorners'

*NAME_MAP

*1 clk
*2 rst
*3 start
*4 x[7]
*5 x[6]
*6 x[5]
*7 x[4]
*8 x[3]
*9 x[2]
*10 x[1]
*11 x[0]
*12 y[7]
*13 y[6]
*14 y[5]
*15 y[4]
*16 y[3]
*17 y[2]
*18 y[1]
*19 y[0]
*20 q[7]
*21 q[6]
*22 q[5]
*23 q[4]
*24 q[3]
*25 q[2]

Plain Text  Tab Width: 8  Ln 1, Col 1  INS
```

6)tempus_script.tcl

```
tempus_script.tcl
~/.asiclab/pd_divider

divder_spfef.spfef  tempus_script.tcl

Open  Save  --  X

set_db init_lib_search_path ./lib/timing

read_libs {fast_vddlv0_basicCells_hvt.lib slow_vddlv0_basicCells_hvt.lib fast_vddlv0_basicCells.lib slow_vddlv0_basicCells.lib fast_vddlv0_basicCells_lvt.lib
slow_vddlv0_basicCells_lvt.lib fast_vddlv2_basicCells_hvt.lib slow_vddlv2_basicCells_hvt.lib fast_vddlv2_basicCells.lib slow_vddlv2_basicCells.lib
fast_vddlv2_basicCells_lvt.lib slow_vddlv2_basicCells_lvt.lib }

read_physical -lef {./lib/lef/gsclib045_tech.lef ./lib/lef/gsclib045_hvt_macro.lef ./lib/lef/gsclib045_macro.lef ./lib/lef/gsclib045_lvt_macro.lef}

read_netlist ./divider_netlist.v
init_design

read_def ./divider_def.def

read_sdc ./divider_sdc.sdc
read_spfef ./divider_spfef.spfef

set_db timing_analysis_type ocv
set_db timing_analysis_cpvr both

#####
# Turn on SI
#####
set_db delaycal_enable_si false
set_db si_glitch_enable_report true
set_db si_delay_separate_on_data true
set_db si_delay_enable_double_clocking_check true
set_db si_delay_enable_report true

#####
# Run timing
#####
update_timing -full

#####
# Run reports
#####
report_timing -split_delay -fields {instance cell arc transition load delay incr_delay }

report_noise -txtfile ./reports/glitch.txt

#check design -type all -out file design.rpt

Tcl  Tab Width: 8  Ln 1, Col 1  INS
```

```
Activities Terminal Nov 7 19:14
23bvd0030@cadence-server:pd_dvldwr

File Edit View Search Terminal Tabs Help

23bvd0030@cadence-server:pd_dvldwr

Required Time:= 0.398
Launch Clock:= 0.160
Data Path:= 0.073
Slack:= 0.225

-----
# Timing Point      Flags Arc Edge Coll Fanout Trans Delay Arrival
#                  (ns) (ns) (ns) (ns) (ns) (ns)
-----
count reg11/CK      - CK->Q F OFFRHOXILVT 33 0.166 - 0.166
count reg11/Q       - CK->Q F OFFRHOXILVT 7 0.100 0.060 0.160
g2498_2802/Y        - B->Y R NAND3XILVT 1 0.022 0.013 0.173
RC CG HIER INST3/en - enable R RC CG MCD 3 - 0.173
RC CG HIER INST3/RC - CG CG INST3/E - E R TLATNTSGAXILVT 1 0.069 0.060 0.173
-----

ntempus 36>
ntempus 36>
ntempus 36>
ntempus 36> #####
ntempus 36> # Run a whole list of common reports
ntempus 40> #####
ntempus 41> file mkdir reports
ntempus 42> source ./reports.tcl>Loading (eight bit divider)
ntempus 42> source ./reports.tcl
**ERROR: [IMPSE 112]: File './reports.tcl': couldn't read file './reports.tcl': no such file or directory.

ntempus 43> report analysis coverage
#####
# Generated by: Cadence Tempus 22.11-s001.1
# OS: Linux x86_64 (Host: 2D cadence-server.vit.ac.in)
# Generated on: Fri Nov 7 19:12:50 2025
# Design: eight-bit divider
# Command: report analysis coverage
#####

TIMING CHECK COVERAGE SUMMARY
-----
Check Type      No. of Met Violated Untested
Checks
-----
ExternalDelay (Early) 17 17 (100%) 0 (0%) 0 (0%)
ExternalDelay (Later) 17 17 (100%) 0 (0%) 0 (0%)
Hold 47 47 (100%) 0 (0%) 0 (0%)
Library Clock Gating Hold 0 4 (50%) 0 (0%) 4 (50%)
Library Clock Gating Setup 0 4 (50%) 0 (0%) 4 (50%)
Pulsewidth 149 162 (88%) 0 (0%) 47 (32%)
Recovery 47 47 (100%) 0 (0%) 0 (0%)
Removal 47 47 (100%) 0 (0%) 0 (0%)
Setup 47 47 (100%) 0 (0%) 0 (0%)
-----

ntempus 44>
```

9) Answer ka questions

Q.1) Was the spef complete?

Yes, the SPEF file was successfully generated, and extraction of RC was also successfully completed.

Q.2) Were the constraints complete?

Yes, all the timing constraints were properly defined and complete.

Q.3) How many clocks were used?

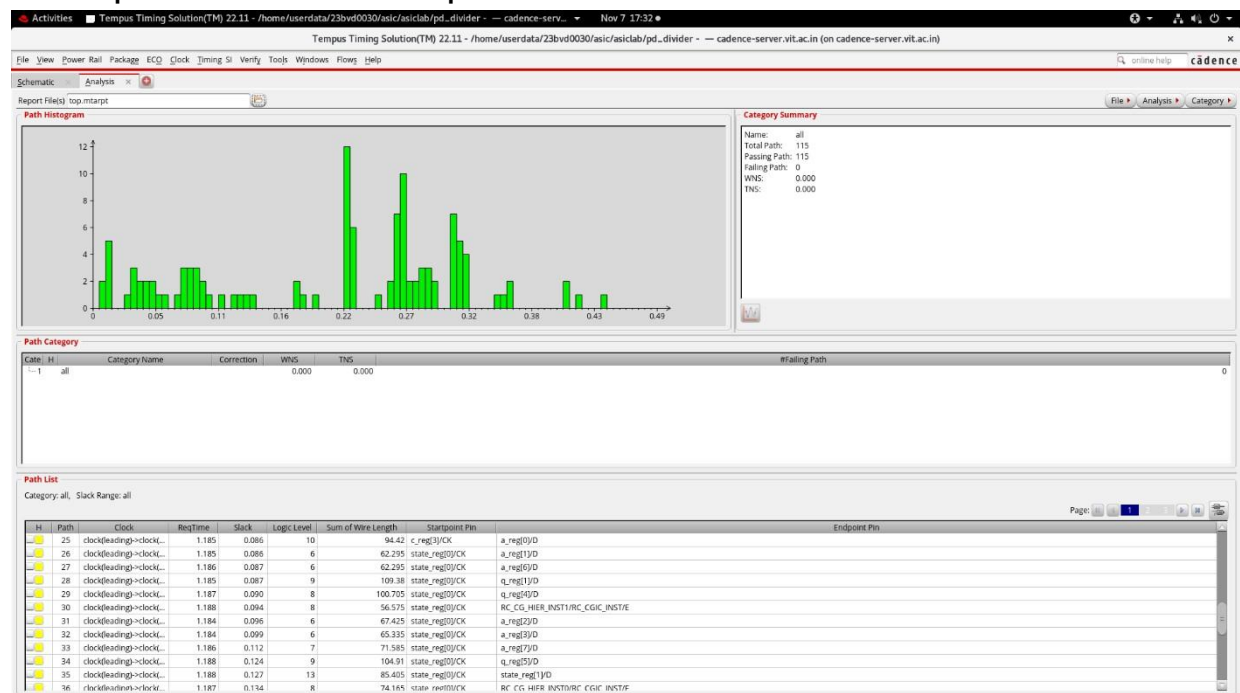
Only one clock is present.

Q.4) Were a lot of paths impacted by constants?

Some of the paths were impacted by the constants.

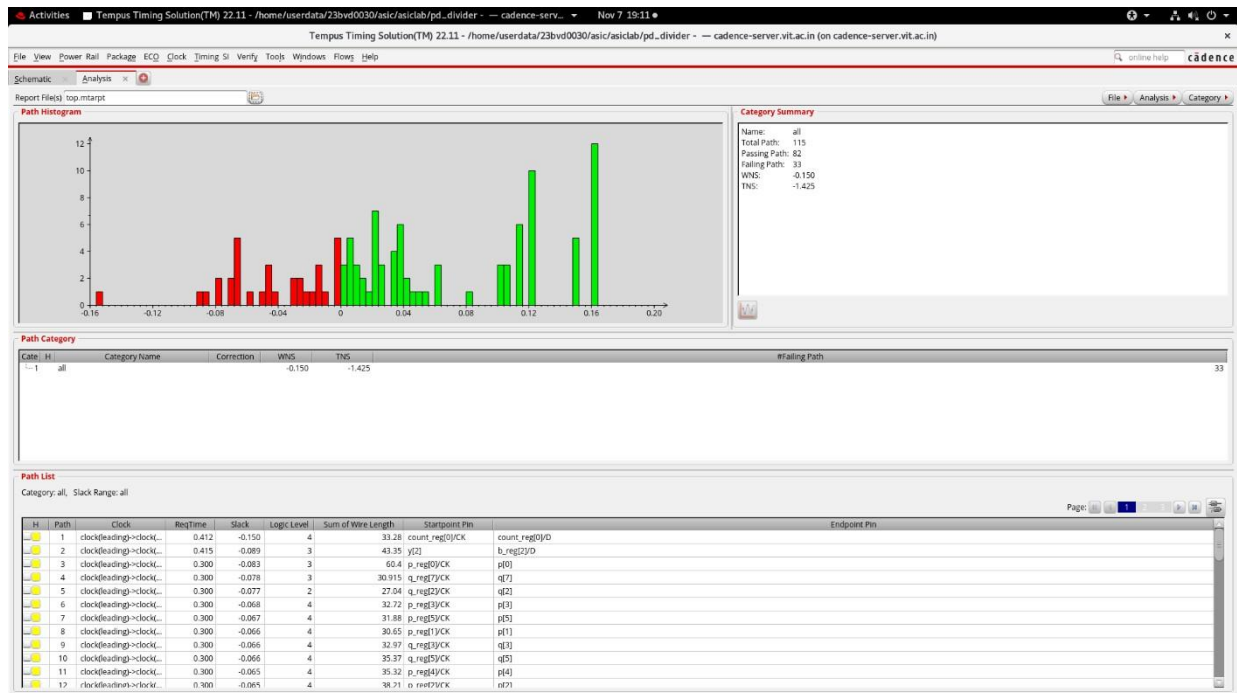
Q.5) How many setup violations are reported?

0 setup violations were reported.



Q.6) Is the worst timing do in inputs or reg-reg paths?

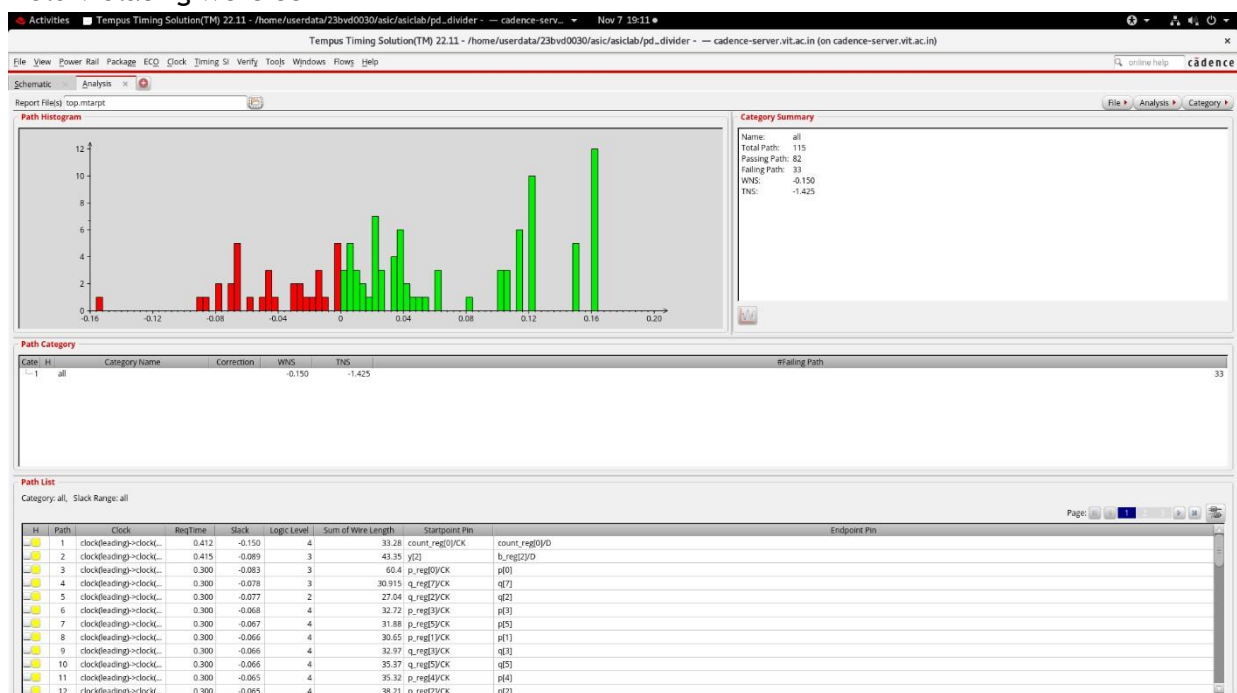
Worst timing path was from a_reg[1]/CK to q_reg[1]/D , hold slack of -0.013.



Q.7) Is there a single Startpoint that causes the timing issues?
 No there is not a single start point that causes the timing issues

Q.8) What is the worst setup and hold violation?
 Worst setup violation : none.
 Worst hold violation: -0.013

10) Iterations
 Setup violating path were 0;
 Hold violating were 33



Made the violations to zero by adding repeaters.

```
tempuszero.tcl
--resic/asiclab/pd_divider

place_default -eco trow -hbar0 -tence -r155
eco_add_repeater -cells BUF20HVT -net n_27 -name FE_ECOC73_n_27
eco_add_repeater -cells BUF20HVT -net FE_PHN142_y_2 -name FE_ECOC74_FE_PHN142_y_2
eco_add_repeater -cells BUF20HVT -net {p[0]} -name FE_ECOC75_p_0
eco_add_repeater -cells BUF20HVT -net {q[7]} -name FE_ECOC76_q_7
eco_add_repeater -cells BUF20HVT -net FE_ESON24_n_106 -name FE_ECOC77_FE_ESON24_n_106
eco_add_repeater -cells BUF20HVT -net FE_ESON61_FE_PHN211_n_94 -name FE_ECOC78_FE_ESON61_FE_PHN211_n_94
eco_add_repeater -cells BUF20HVT -net FE_ESON25_n_83 -name FE_ECOC79_FE_ESON25_n_83
eco_add_repeater -cells BUF20HVT -net FE_PHN207_n_25 -name FE_ECOC80_FE_PHN207_n_25
eco_add_repeater -cells BUF20HVT -net RC.CG.HIER_INST2/FE_ESON56_n_9 -name FE_ECOC81_FE_ESON56_n_9
eco_add_repeater -cells BUF20HVT -net FE_ESON49_n_104 -name FE_ECOC82_FE_ESON49_n_104
eco_add_repeater -cells BUF20HVT -net FE_ESON66_FE_PHN159_n_96 -name FE_ECOC83_FE_ESON66_FE_PHN159_n_96
eco_add_repeater -cells BUF20HVT -net FE_ESON39_n_88 -name FE_ECOC84_FE_ESON39_n_88
eco_add_repeater -cells BUF20HVT -net FE_PHN147_FE_OCPN72_n_9 -name FE_ECOC85_FE_PHN147_FE_OCPN72_n_9
eco_add_repeater -cells BUF20HVT -net FE_PHN147_FE_OCPN72_n_9 -name FE_ECOC86_FE_PHN147_FE_OCPN72_n_9
eco_add_repeater -cells BUF20HVT -net FE_ESON12_n_37 -name FE_ECOC87_FE_ESON12_n_37
eco_add_repeater -cells BUF20HVT -net FE_ESON72_FE_PHN144_y_1 -name FE_ECOC88_FE_ESON72_FE_PHN144_y_1
eco_add_repeater -cells BUF20HVT -net FE_ECON88_FE_ESON72_FE_PHN144_y_1 -name FE_ECOC89_FE_ESON72_FE_PHN144_y_1
eco_add_repeater -cells BUF6LVT -net FE_PHN134_y_4 -name FE_ECOC90_FE_PHN134_y_4
eco_add_repeater -cells BUF6LVT -net FE_ECON90_FE_PHN134_y_4 -name FE_ECOC91_FE_PHN134_y_4
eco_add_repeater -cells BUF6LVT -net FE_ESON47_n_100 -name FE_ECOC92_FE_ESON47_n_100
eco_add_repeater -cells BUF20HVT -net {q[6]} -name FE_ECOC93_q_6
eco_add_repeater -cells BUF20HVT -net RC.CG.HIER_INST3/FE_ESON46_n_116 -name FE_ECOC94_FE_ESON46_n_116
eco_add_repeater -cells BUF20HVT -net RC.CG.HIER_INST3/FE_ESON46_n_116 -name FE_ECOC95_FE_ESON46_n_116
eco_add_repeater -cells BUF20HVT -net {q[6]} -name FE_ECOC96_q_6
eco_add_repeater -cells BUF20HVT -net FE_ECON92_FE_ESON47_n_100 -name FE_ECOC97_FE_ESON47_n_100
```

Write_eco -format innovus tempuszero.tcl

Innovus:

After opening post_route.dat;

source tempuszero.tcl

place_eco

route_eco

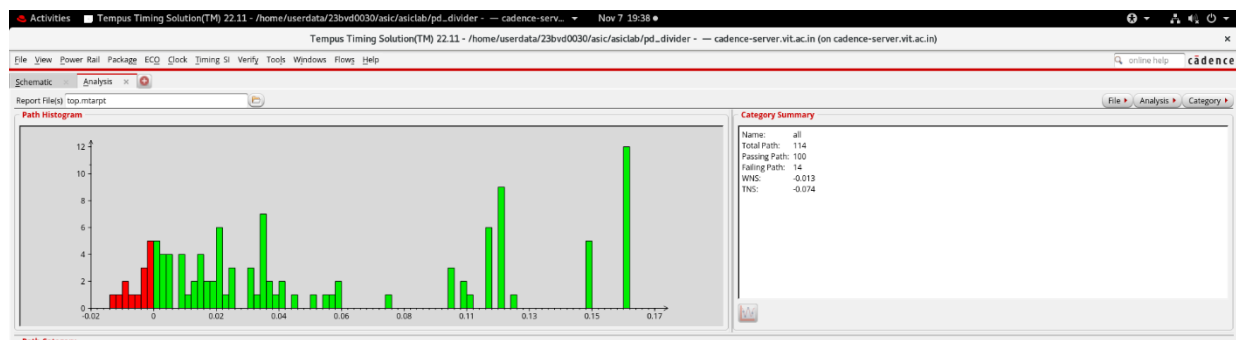
Then save by write_db DBS /final.dat

After iteration 1 of fixing violations and generating netlist, spef, sdc and def file in Innovus we take these files into Tempus to check for any other violations that may have occurred while placement or routing.

Tempus

Setup Violations = 0

Hold Violations = 14



11) tempuszero2.tcl file

```
eco_add_repeater -cells BUF20HVT -net {p[0]} -name FE_ECOC100_p_0
eco_add_repeater -cells BUF20HVT -net {q[4]} -name FE_ECOC101_q_4
eco_add_repeater -cells BUF20HVT -net {q[1]} -name FE_ECOC102_q_1
eco_add_repeater -cells BUF20HVT -net {q[0]} -name FE_ECOC103_q_0
eco_add_repeater -cells BUF20HVT -net done -name FE_ECOC104_done
eco_add_repeater -cells BUF20HVT -net {q[7]} -name FE_ECOC105_q_7
eco_add_repeater -cells DLY1X1 -net {p[7]} -name FE_ECOC106_p_7
eco_add_repeater -cells DLY1X1 -net FE_ECON74_FE_PHN142_y_2 -name FE_ECOC107_FE_PHN142_y_2
eco_add_repeater -cells DLY1X1 -net {q[4]} -name FE_ECOC108_q_4
eco_add_repeater -cells DLY1X1 -net {q[2]} -name FE_ECOC109_q_2
eco_add_repeater -cells DLY1X1 -net FE_ECON73_n_27 -name FE_ECOC110_n_27
eco_add_repeater -cells DLY1X1 -net FE_ECON73_n_27 -name FE_ECOC111_n_27
eco_add_repeater -cells DLY1X1 -net {p[3]} -name FE_ECOC112_p_3
eco_add_repeater -cells DLY1X1 -net {p[5]} -name FE_ECOC113_p_5
eco_add_repeater -cells DLY1X1 -net {p[1]} -name FE_ECOC114_p_1
eco_add_repeater -cells DLY1X1 -net {p[1]} -name FE_ECOC115_p_1
eco_add_repeater -cells DLY1X1 -net {q[3]} -name FE_ECOC116_q_3
eco_add_repeater -cells DLY1X1 -net {q[3]} -name FE_ECOC117_q_3
eco_add_repeater -cells DLY1X1 -net {q[3]} -name FE_ECOC118_q_3
```

Innovus:

After opening post_route;
source tempuszero2.tcl
place_eco
route_eco

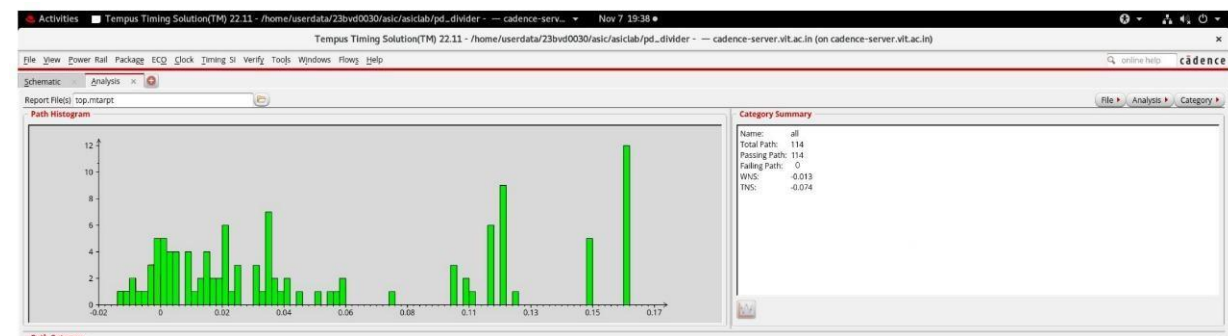
Then save by write_db DBS /final.dat

#The above commands were used to place and route the changes and addition of repeaters in tempus

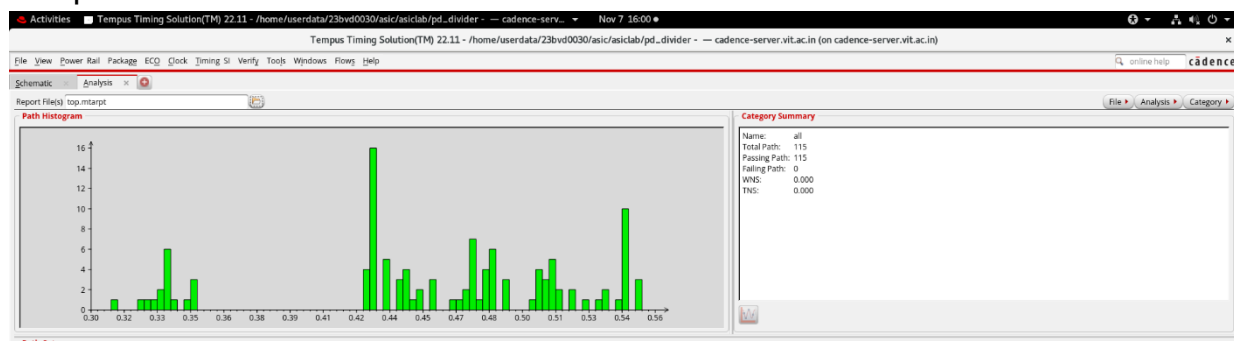
#After iteration 2 of fixing violations and generating new netlist, spef, sdc and def file in Innovus we take these files into Tempus to check for any other violations that may have occurred while placement or routing.

Tempus

Hold Violations=0



Setup Violations = 0



13) report_analysis_coverage

```
Nov 7 19:14
23bvd0030@cadence-server:pd_divider

# Timing Point      Flags Arc Edge Cell Fanout Trans Delay Arrival
#-----
count reg[3]/CK      - CK  R (arrival) 13 0.100 - 0.100
count reg[3]/Q      - CK-Q F DIFFNOX1LVT 7 0.100 0.060 0.160
g2498 2802/Y      - B-Y R NAND3X1LVT 1 0.022 0.013 0.173
RC CG HIER INST3/enable enable R RC CG MOD 3 - - 0.173
RC CG HIER INST3/RC CG IC INST/E E R TLATNTSCAX3LVT 1 0.060 0.060 0.173

#-----
#tempus 36-
#tempus 36-
#tempus 36-
#tempus 36- *****
#tempus 39- # Run a whole list of common reports
#tempus 40- *****
#tempus 41- file mkdir reports
#tempus 42- source ./reports.tcl>Loading (eight bit divider)
#tempus 43- source ./reports.tcl
**ERROR: (IMPSE-112): File './reports.tcl': couldn't read file './reports.tcl': no such file or directory.

#tempus 43- report analysis coverage
*****
# Generated by: Cadence Tempus 22.11-9001.1
# OS: Linux x86_64(Host ID cadence-server.vit.ac.in)
# Generated on: Fri Nov 7 19:11:50 2025
# Design: eight bit divider
# Command: report analysis coverage
*****
TIMING CHECK COVERAGE SUMMARY
-----
Check Type      No. of Met Violated Untested
Checks
-----
ExternalDelay (Early) 17 17 (100%) 0 (0%) 0 (0%)
ExternalDelay (Late) 17 17 (100%) 0 (0%) 0 (0%)
Hold 47 47 (100%) 0 (0%) 0 (0%)
Library Clock Gating Hold 8 4 (50%) 0 (0%) 4 (50%)
Library Clock Gating Setup 6 4 (50%) 0 (0%) 4 (50%)
Pulsewidth 140 102 (80%) 0 (0%) 47 (31%)
Recovery 47 47 (100%) 0 (0%) 0 (0%)
Removal 47 47 (100%) 0 (0%) 0 (0%)
Setup 47 47 (100%) 0 (0%) 0 (0%)
-----
#tempus 44-
```

14) check drc violations and connectivity

```
@innovus 2> check_drc
#-check_same_via_cell true # bool, default=false, user setting
*** Starting Verify DRC (MEM: 1710.8) ***

VERIFY DRC ..... Starting Verification
VERIFY DRC ..... Initializing
VERIFY DRC ..... Deleting Existing Violations
VERIFY DRC ..... Creating Sub-Areas
VERIFY DRC ..... Using new threading
VERIFY DRC ..... Sub-Area: {0.000 0.000 57.600 57.380} 1 of 1
VERIFY DRC ..... Sub-Area : 1 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:00.1 ELAPSED TIME: 0.00 MEM: 275.1M) ***

@innovus 3> check_connectivity
VERIFY_CONNECTIVITY use new engine.

***** Start: VERIFY CONNECTIVITY *****
Start Time: Fri Nov 7 22:03:39 2025

Design Name: eight_bit_divider
Database Units: 2000
Design Boundary: (0.0000, 0.0000) (57.6000, 57.3800)
Error Limit = 1000; Warning Limit = 50
Check all nets

Begin Summary
Found no problems or warnings.
End Summary

End Time: Fri Nov 7 22:03:39 2025
Time Elapsed: 0:00:00.0

***** End: VERIFY CONNECTIVITY *****
Verification Complete : 0 Viols. 0 Wrngs.
(CPU Time: 0:00:00.0 MEM: 0.000M)
```

15)Inferences:

Physical implementation using Cadence Innovus and timing verification using Cadence Tempus were completed with STA analysis. Complete RC information was obtained from the SPEF file for accurate delay modeling. Application of all the constraints was done, and setup and hold check analyses were performed on the design for all paths. Initially, some violations were found that were corrected systematically by performing timing optimizations.

After re-analysis, the design achieved zero setup and hold violations, which confirmed that it met all the timing requirements under the defined operating conditions. This assignment gave practical understanding of timing closure flow, handling constraints, and the integration between Innovus and Tempus for post-layout STA.